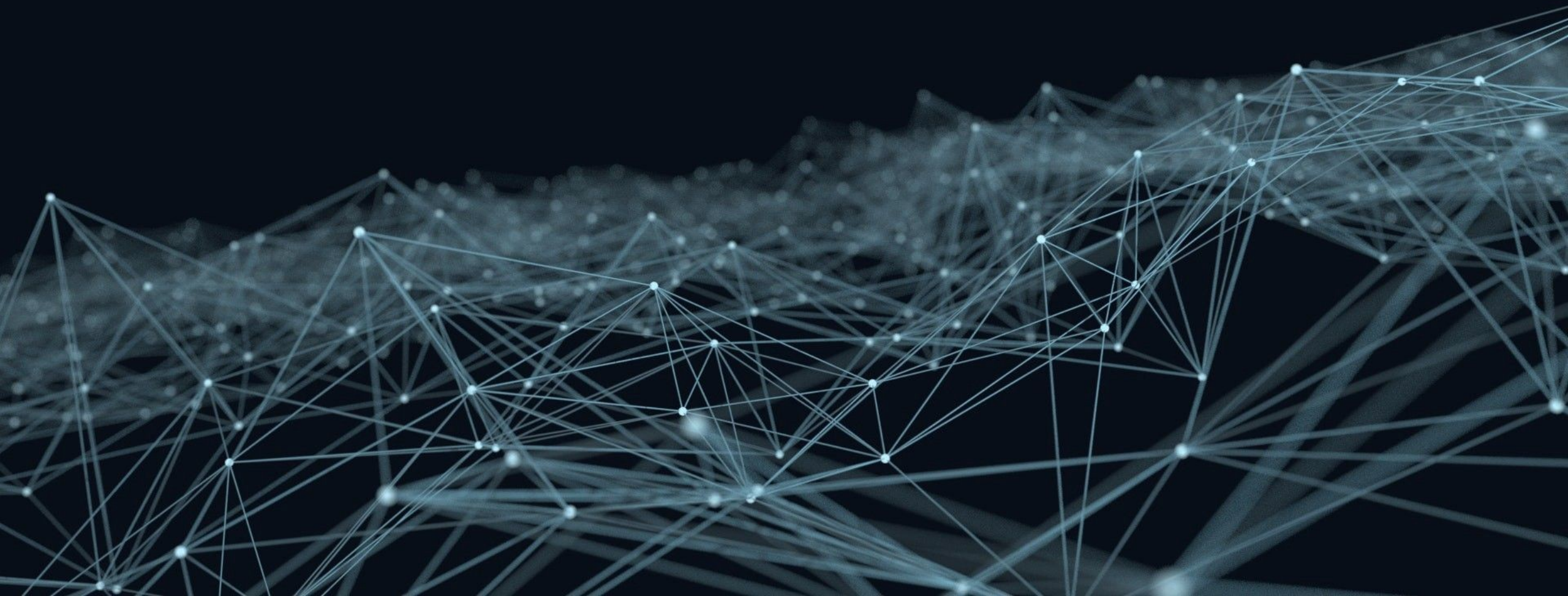
Java Full Stack Development

2. Oauth Hardening



Our Stack

- Our labs
  - Auth server
    - In-memory users
    - Hardcoded client secrets
    - RSA key generated fresh each startup
  - Resource server
    - Validates JWTs against the auth server's JWKS
    - No audience validation

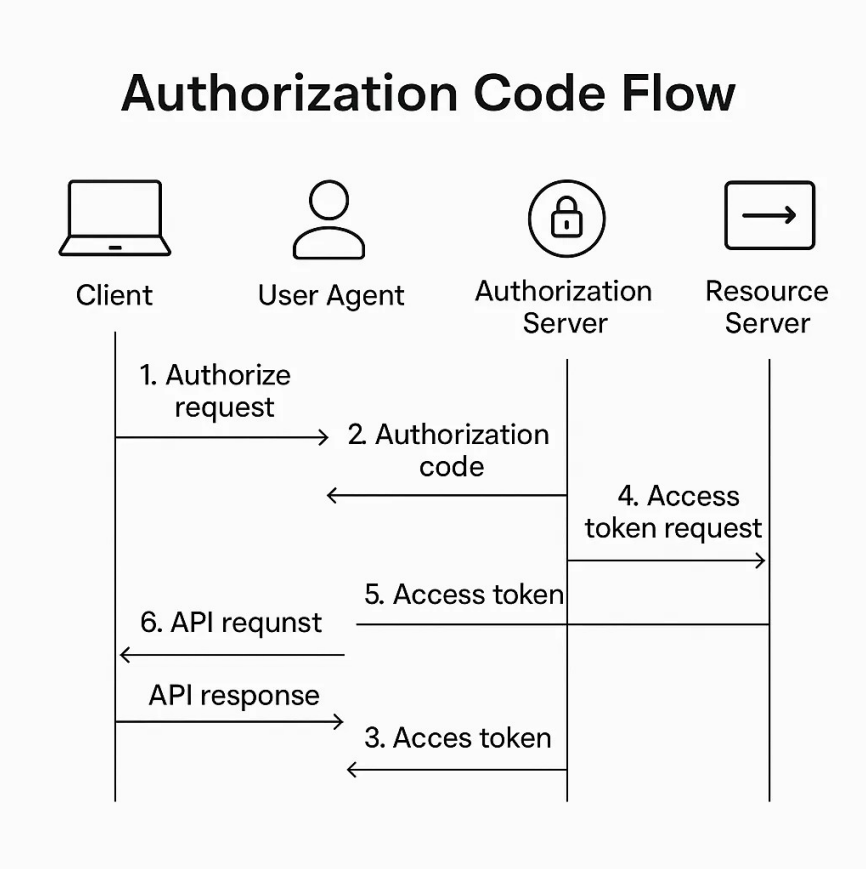
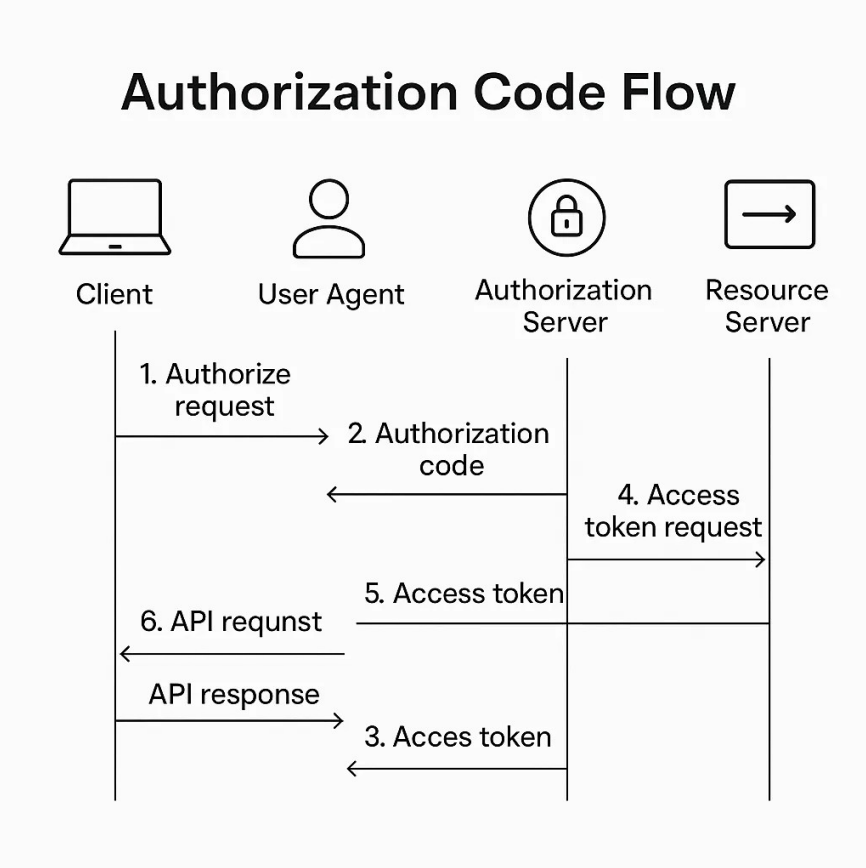


Image Credit: https://ssojet.com/blog/oauth-2-0-security-best-practices-how-to-secure-oauth-tokens-and-why-use-pkce/

Our Stack

- Our labs
  - BFF
    - Stores tokens in HTTP session, default cookie attributes, default token timeouts
  - All three
    - Running on localhost
    - No HTTPS
    - No rate limits
    - No audit logging



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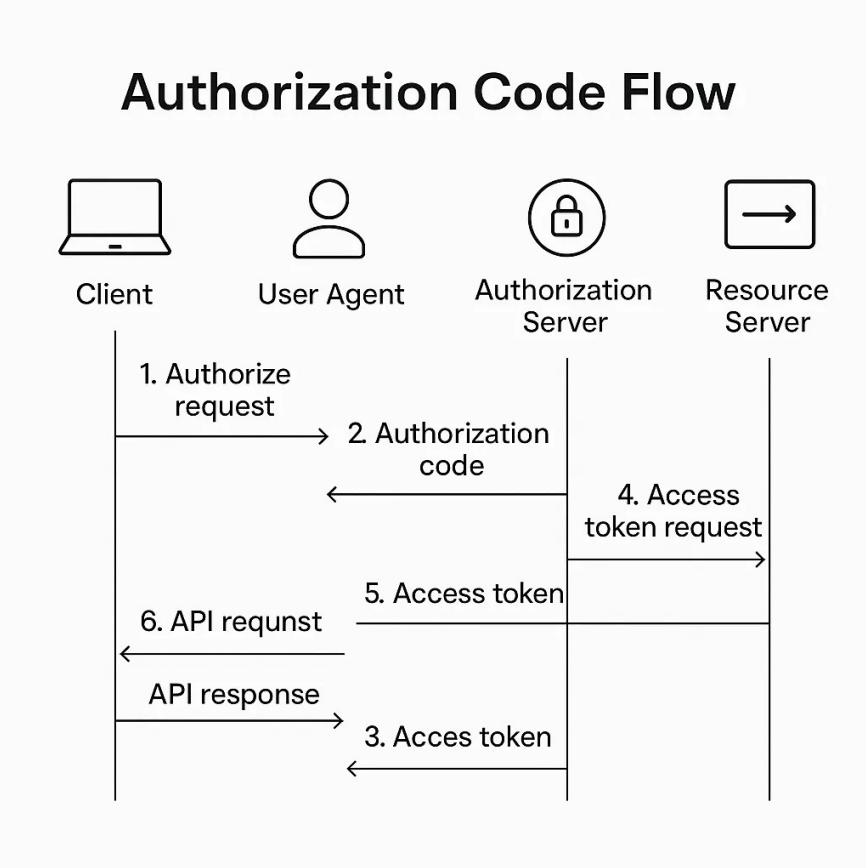
Our Stack

- Auth server compromises
  - The auth server uses in-memory users
    - Alice, bob with hardcoded passwords
  - It uses {noop} password encoding which Spring explicitly warns against in production
  - It generates a fresh RSA key pair on every startup, which means every restart invalidates all previously-issued tokens
  - None of these are appropriate for production
  - But these options kept the lab simpler.



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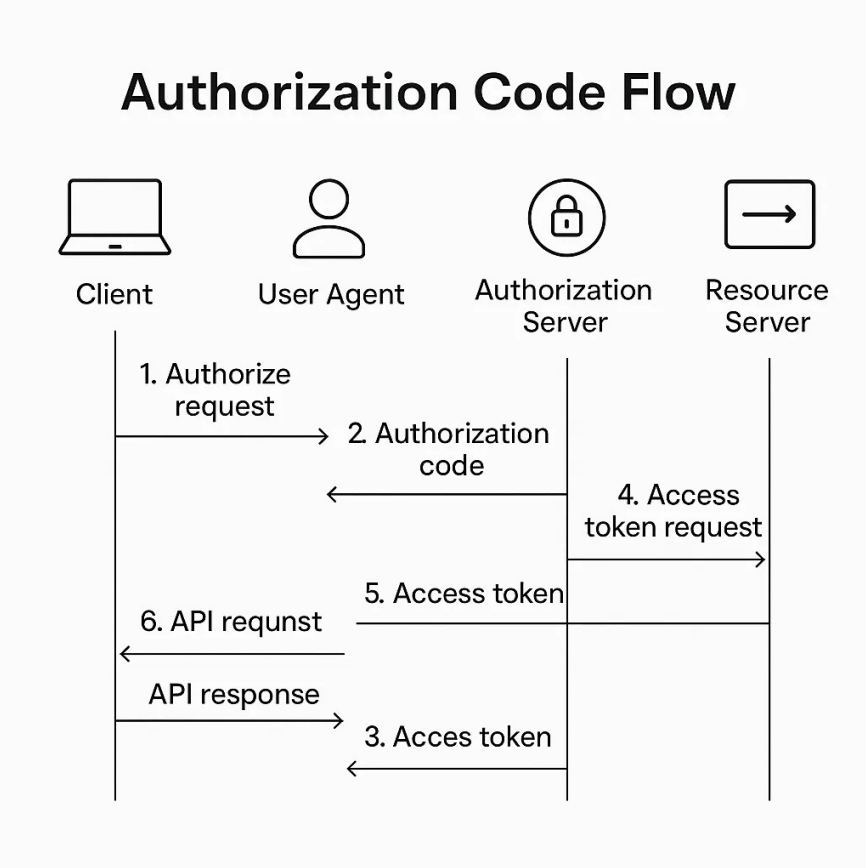
Our Stack

- Resource server compromises
  - The bankserver validates JWT signatures against the auth server's JWKS endpoint and checks that the issuer claim matches
  - It does not validate the audience claim
    - This means a token issued for the bankbff client could theoretically be replayed against a different resource server that trusts the same auth server
    - A security gap that audience validation would close.



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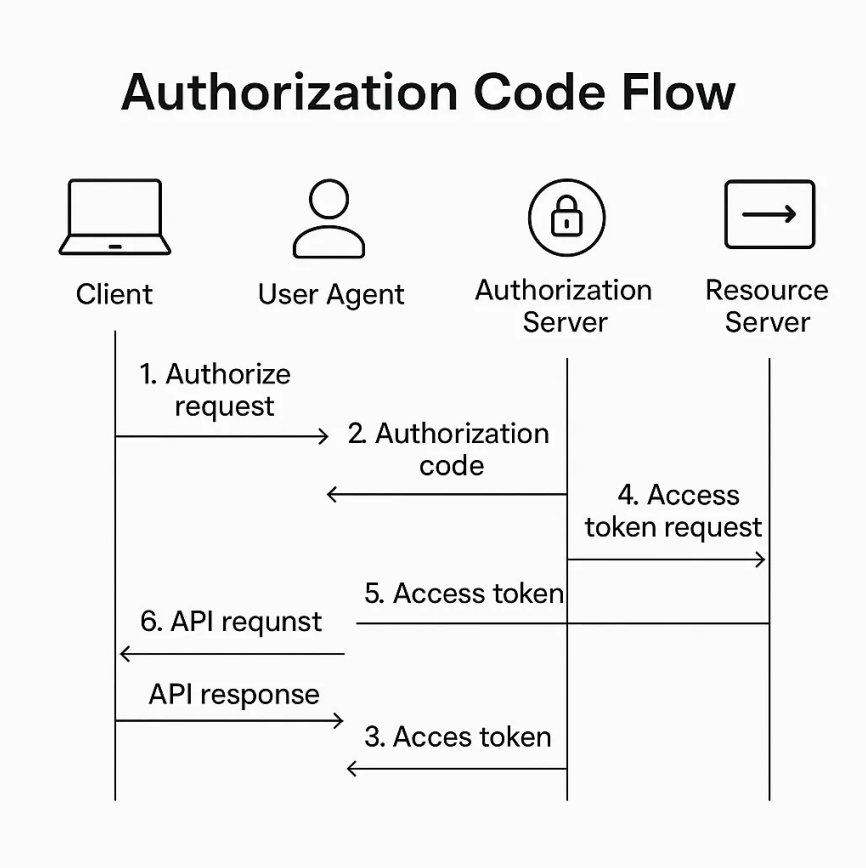
Our Stack

- BFF compromises
  - The bankbff stores OAuth tokens in the HTTP session
    - Which is the default Spring Security behavior.
  - The session cookie is HttpOnly (correct)
  - But uses default settings for everything else
    - No Secure flag which is acceptable on localhost, wrong for production
  - SameSite=Lax by default
    - Sometimes correct, sometimes wrong, but always worth being explicit
    - Token lifetimes use the auth server's default of 60 minutes for access tokens, 24 hours for refresh tokens



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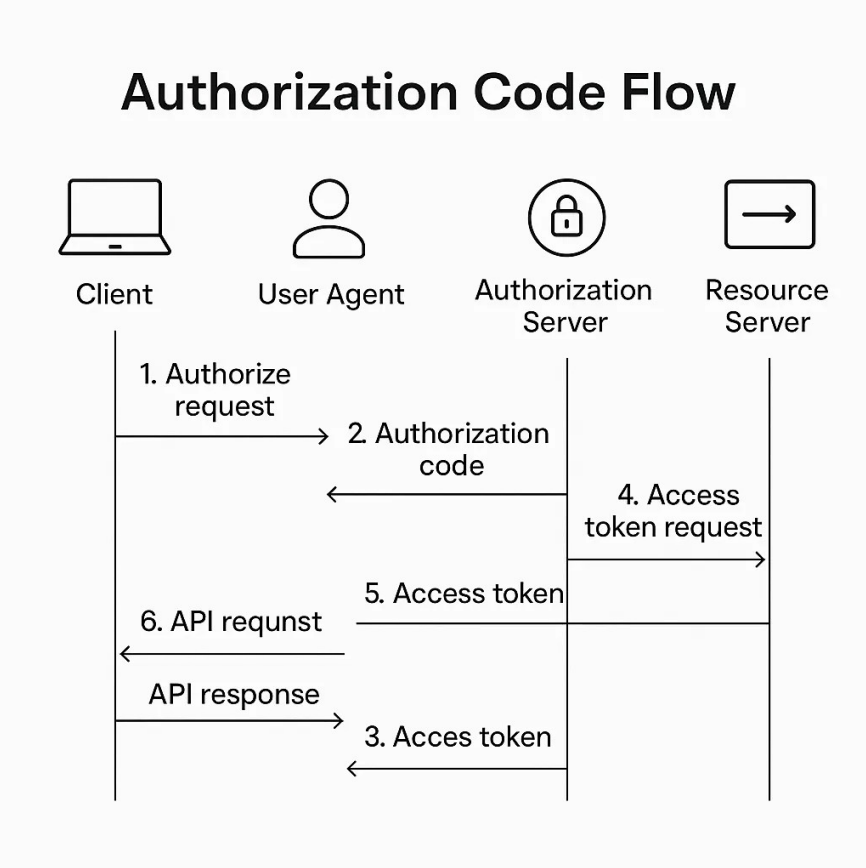
Our Stack

- Cross-cutting compromises
  - Everything runs on http://localhost
  - There is no HTTPS, so all traffic including cookies and tokens is in plaintext.
  - There are no rate limits
    - So an attacker could brute-force credentials or pound the token endpoint
  - There is no audit logging of security-relevant events



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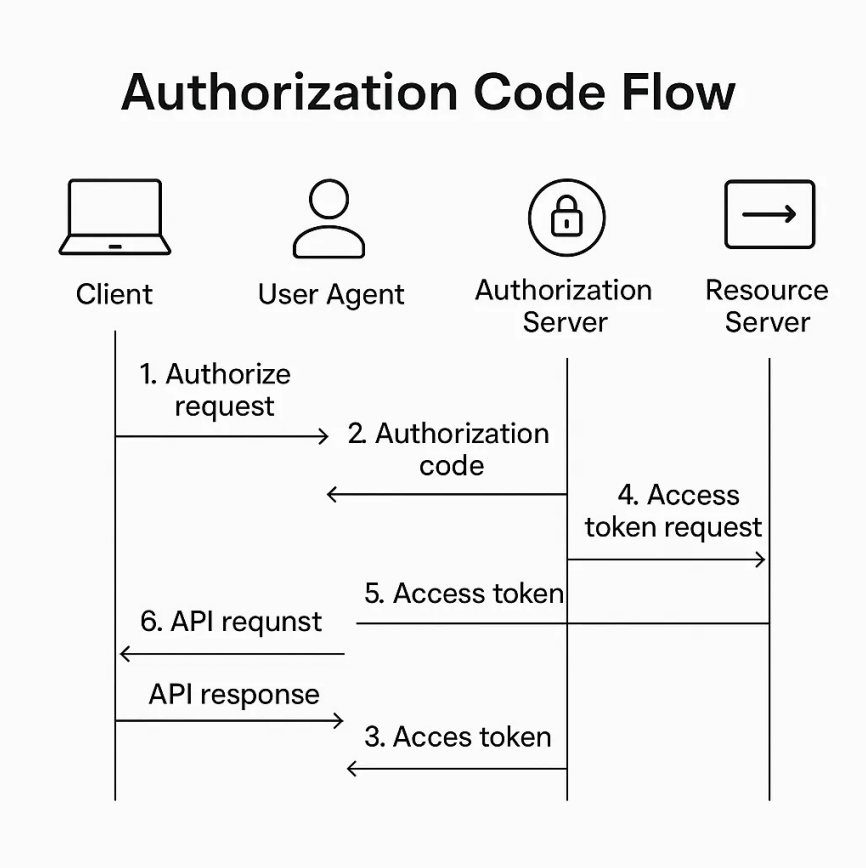
Token Lifetime Tuning

- How long should tokens live?
  - Access tokens
    - Short, minutes, not hours
    - 5-15 minutes is typical for high-security applications
  - Refresh tokens
    - Longer, hours to days, depending on threat model
  - ID tokens:
    - Short, typically same as access tokens
    - Used immediately at login then discarded
- Both rotatable
  - When an access token expires
  - Refresh produces a new pair
  - New access AND new refresh
- Trade-off
  - Shorter is more secure (smaller window if leaked) but more chatty (more refresh calls)



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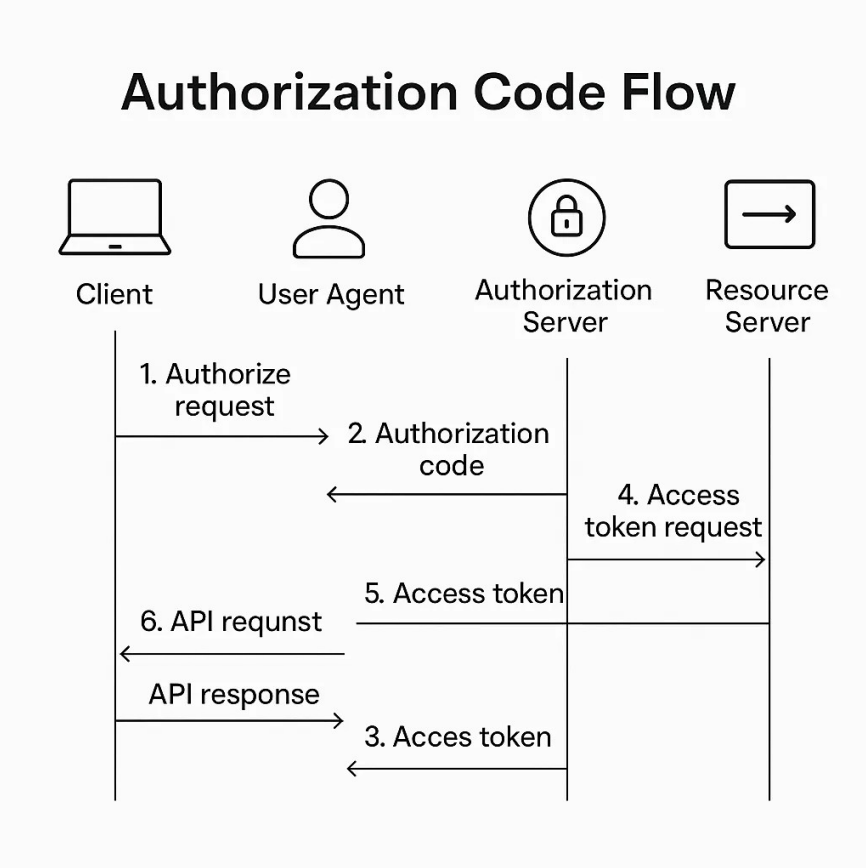
Scope Minimization

- Request only what you need
  - Scopes describe what the application can do, not what the user can do
  - Each OAuth client should request the smallest set of scopes that lets it function
    - The user grants scopes
    - The application receives a token with only those scopes
  - A token cannot grant more authority than its scopes allow
  - Naming convention
    - Scopes describe capabilities (read:accounts, write:transactions)
    - Not roles (HR_MANAGER)



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Roles, Scopes, and Permissions

- Confusing these is the most common OAuth implementation mistake
- Each layer is checked independently
- A request must pass
  - Token has scope
  - User has role
  - Permission allows action

Title: Three Concepts, Three Layers

Concept	Lives at	Describes	Example
Scope	OAuth token	What the application can do	`read:accounts`, `write:transactions`
Role	User identity	What the user IS	`HR_MANAGER`, `VIEWER`, `ADMIN`
Permission	Application	What the user can do in this app	"Can transfer up to $10,000"

Roles, Scopes, and Permissions

- Scopes
  - Are properties of the OAuth token
  - The auth server issues tokens with a fixed set of scopes
    - The ones the user granted
    - Intersected with the ones the client is allowed to request
  - A token's scopes describe what the application can do on behalf of the user
    - Not what the user can do.

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Roles, Scopes, and Permissions

- Roles
  - Are properties of the user's identity
  - They're typically stored in the application's database or fetched from an identity provider
  - A role describes what the user is in the application's authorization model
    - HR_MANAGER, VIEWER, ADMIN, etc.
  - The same user has the same roles regardless of which application they're using

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Roles, Scopes, and Permissions

- Permissions
  - Are application-specific authorization decisions
  - They typically combine roles, business rules, and runtime context
    - "the user has the MANAGER role”
    - AND “the requested transfer amount is under $10,000”
    - AND “it's during business hours, so the action is allowed"
    - Permissions are computed by the application, not granted by the auth server

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Decision Flow

- The decision flow at request time
  - Authentication
    - The token is valid and the user is who they claim to be
  - Scope check
    - The token has the scope required for this API endpoint
  - Role check
    - The user has the role required to even attempt this action
  - Permission check
    - The action is allowed for this specific user given the current context
  - All four must pass for the request to succeed
    - Each layer is a separate check

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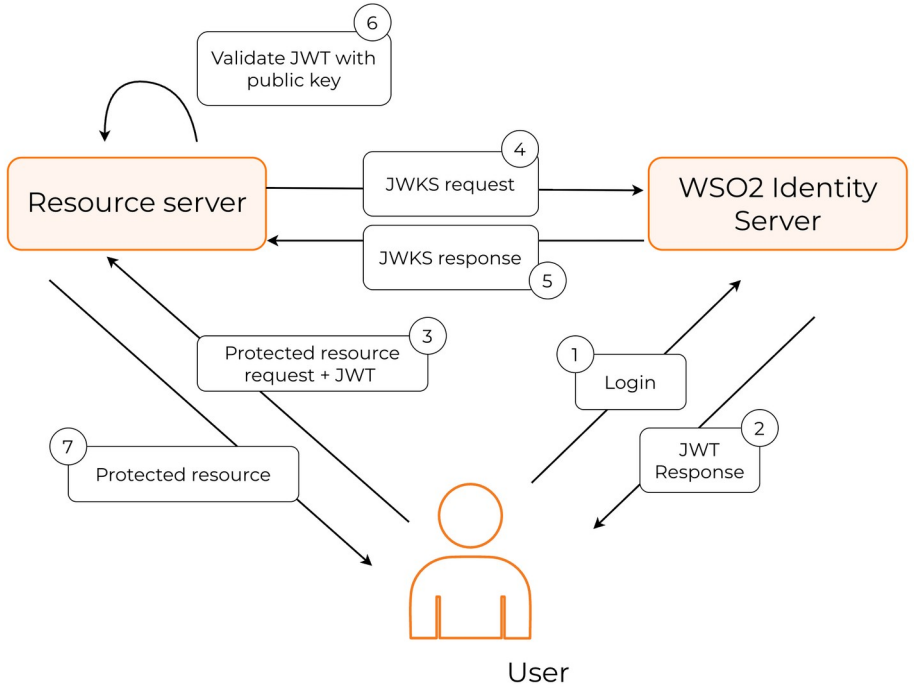
Key Rotation

- JWT verification trustworthiness
  - JWKS = JSON Web Key Set
    - A JSON document listing the public keys the auth server uses to sign tokens
  - Resource servers fetch JWKS at startup and cache it
  - The auth server periodically rotates its signing keys
    - Old keys remain in JWKS during a grace period
  - Spring Security caches JWKS and refreshes it when an unknown kid appears
  - Production-critical
    - Do not write code that hardcodes specific keys



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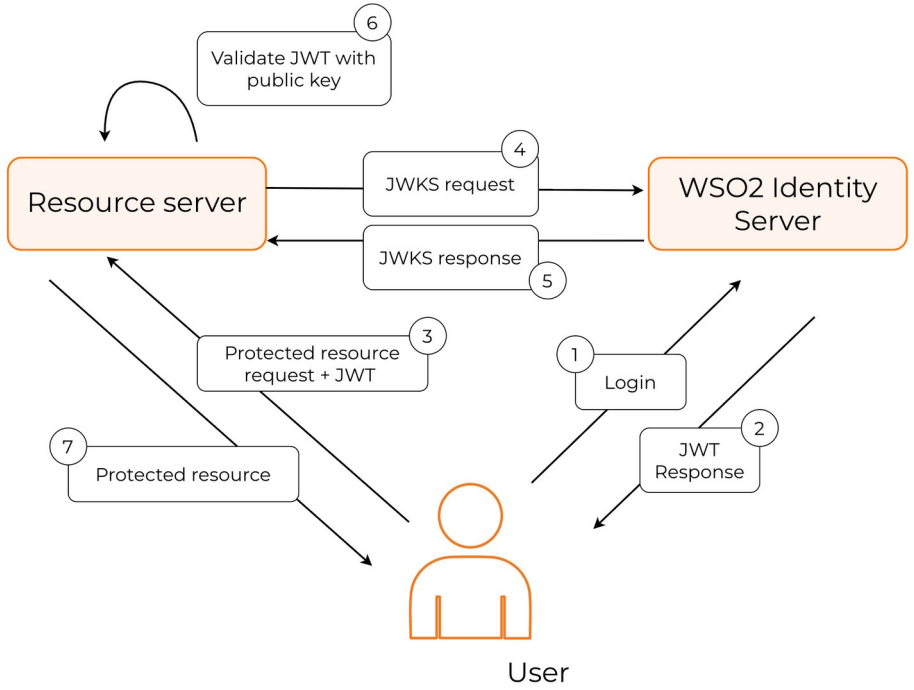
Clock Skew Tolerance

- Server clock mismatch
  - JWTs include iat (issued at), exp (expires at), nbf (not before) all are timestamps
  - Different servers' clocks drift slightly; even with NTP, a few seconds of disagreement is normal
    - Without tolerance, a token issued at 12:00:00 by the auth server might be rejected at 11:59:58 by the resource server
  - Spring Security default
    - 60 seconds of clock skew tolerance; configurable per resource server
  - Production-critical for distributed systems
  - Sever saying:
    - "I'll accept tokens up to N seconds outside my view of valid time"



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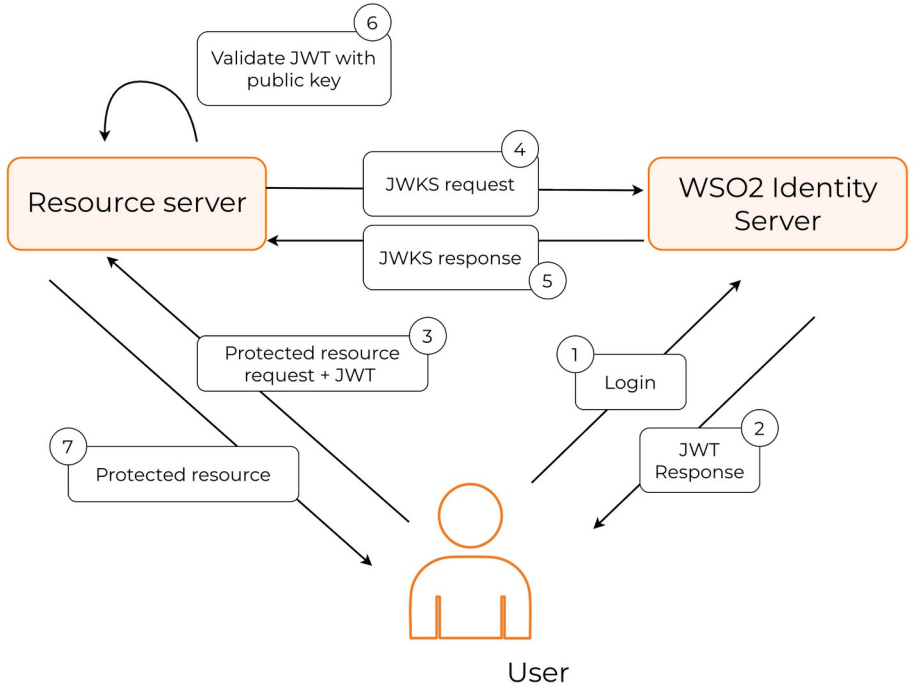
Audience Validation

- The aud claim
  - In a JWT identifies which resource server(s) the token is intended for
  - A resource server should reject tokens where aud does not include itself
  - Without audience validation
    - A token issued for resource server A could be replayed against resource server B
    - If both trust the same auth server
  - This is a security boundary, not just metadata



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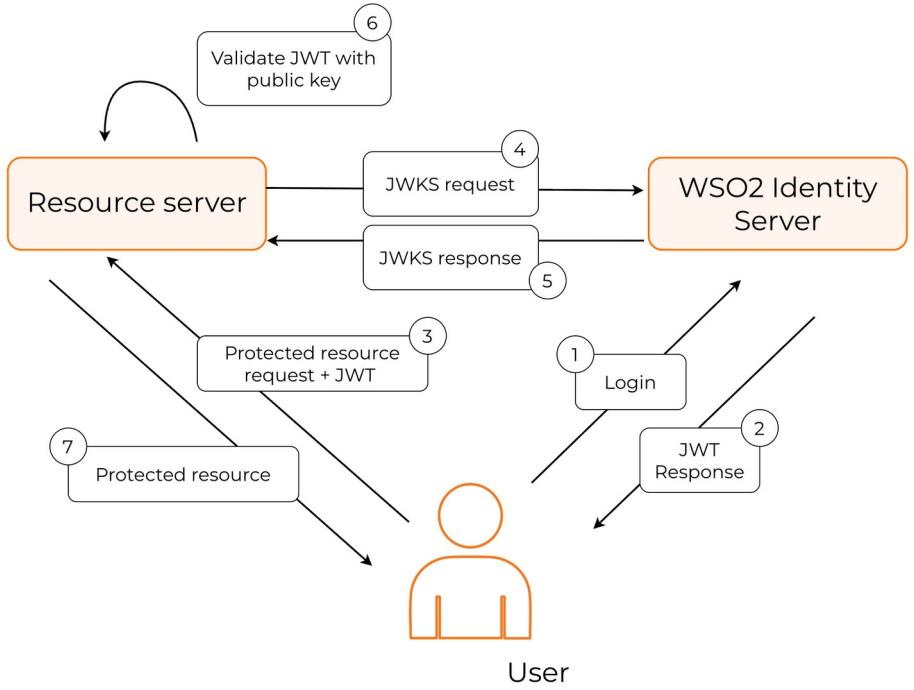
Validating Tokens

- Token Introspection vs. JWT Verification
  - JWT verification
    - Resource server validates the signature using cached JWKS
    - No call to auth server per request
    - Fast: pure local operation, no network call
    - Stale: revoked tokens remain valid until they expire
- Token introspection
  - Resource server calls the auth server's /introspect endpoint for every token
    - Fresh: revocation takes effect immediately
    - Slow: extra network call on every API request
  - The choice depends on threat model
    - How fast must revocation propagate?



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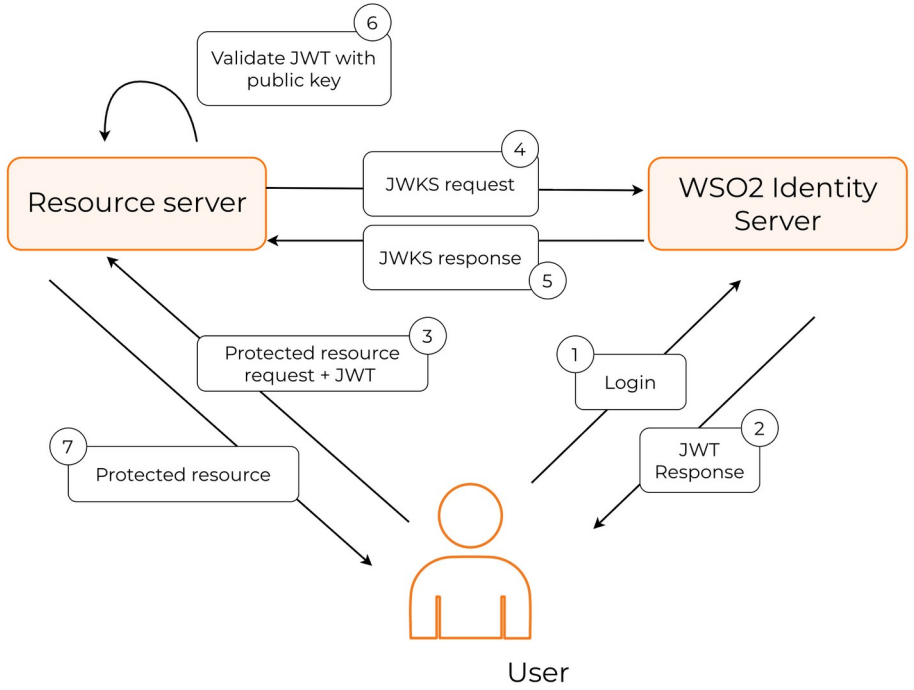
Validating Tokens

- JWT verification
  - Token is self-contained
    - It includes the user identity, the scopes, the expiry, and a signature
    - The resource server verifies the signature against the cached JWKS and trusts everything inside the token until it expires
    - No call to the auth server per request, this is fast and scales well.
  - The downside
    - Revocation is delayed
    - If a user's account is suspended, their existing tokens remain valid until they expire naturally
    - With longer-lived tokens, the revocation gap can be hours



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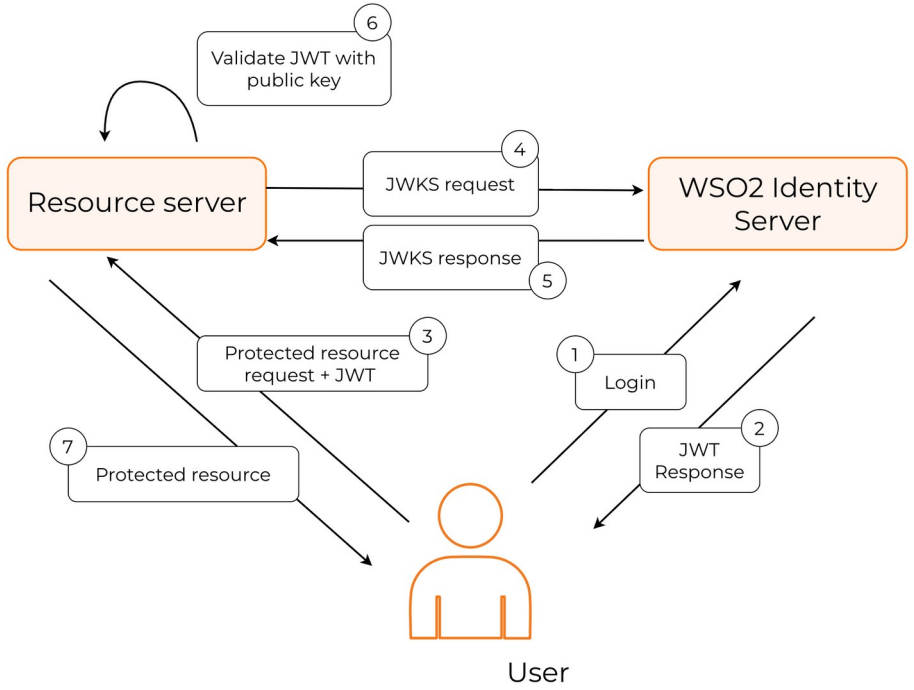
Validating Tokens

- Token introspection
  - The resource server calls the auth server's /introspect endpoint for every API request, sending the token
  - The auth server checks its current state
    - Is this token still valid?
    - Has the user been suspended?
    - Has the application been disabled?
  - Returns either "active" or "inactive"
  - The resource server uses the response to decide whether to allow the request.
  - The benefit
    - Revocation is immediate
    - Suspend the user, and within milliseconds their next API call fails



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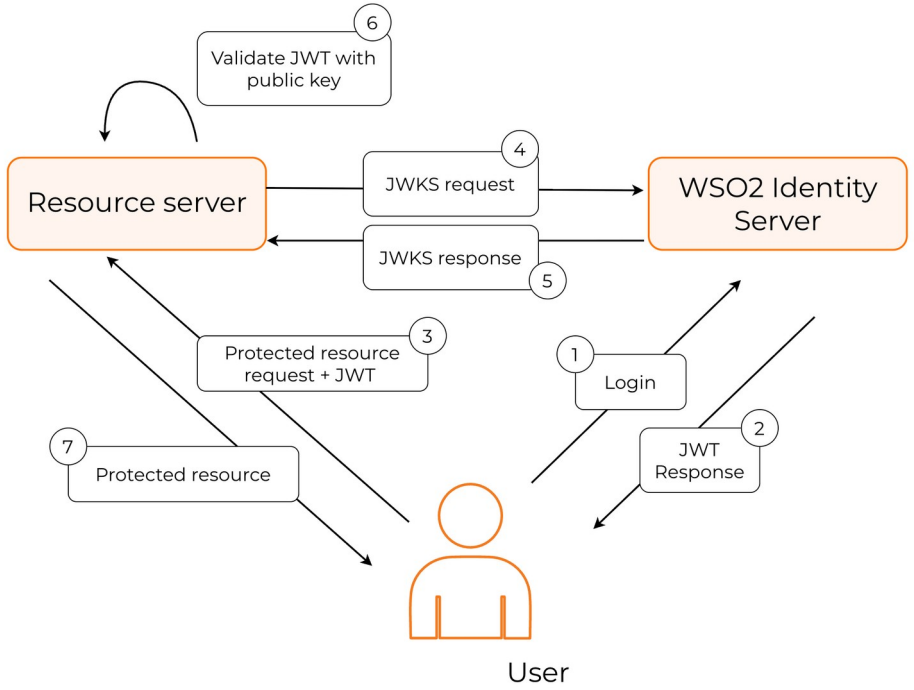
Confused Deputy

- The confused deputy problem
  - A "confused deputy" is a privileged service that performs actions on behalf of less-privileged callers
  - Without scope checks
    - A caller could ask the deputy to do anything the deputy is allowed to do
  - OAuth scopes solve this by attaching the caller's intended capabilities to the request
  - The deputy enforces scope
    - It only does what the token's scopes allow, not what the deputy itself can do



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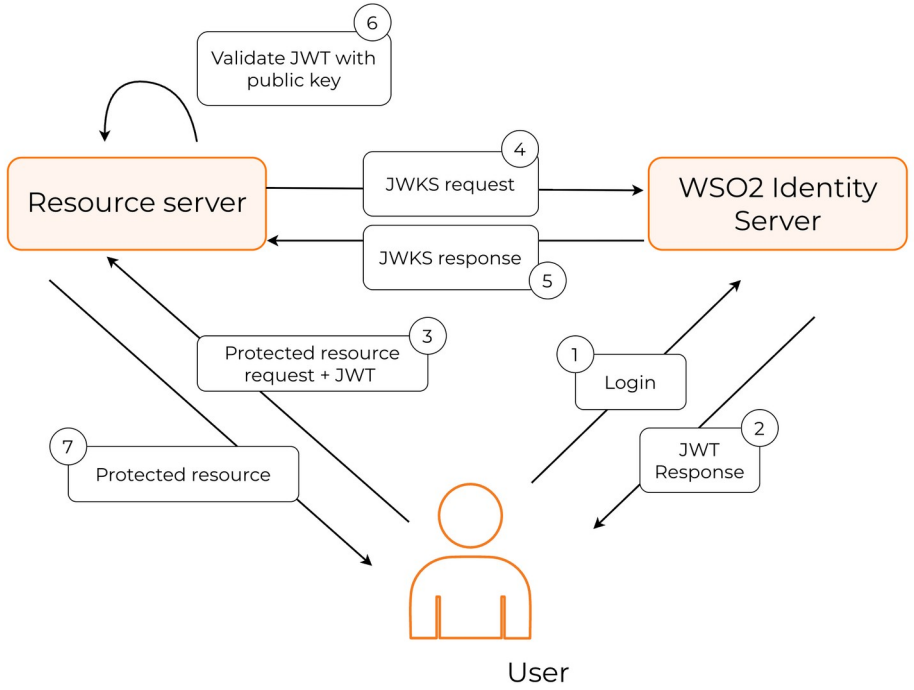
Cookies

- Session management and cookies
  - Session timeout
    - how long an idle session remains valid (15-30 minutes for banking, longer for consumer apps)
  - Sliding vs. absolute expiry
    - Sliding resets on each request, absolute expires regardless of activity
  - Cookie attributes
    - HttpOnly, Secure, SameSite
    - Each closes a different attack vector
  - Concurrent session control
    - How many simultaneous sessions per user
  - Session fixation protection
    - Rotate session ID on login (Spring Security default)



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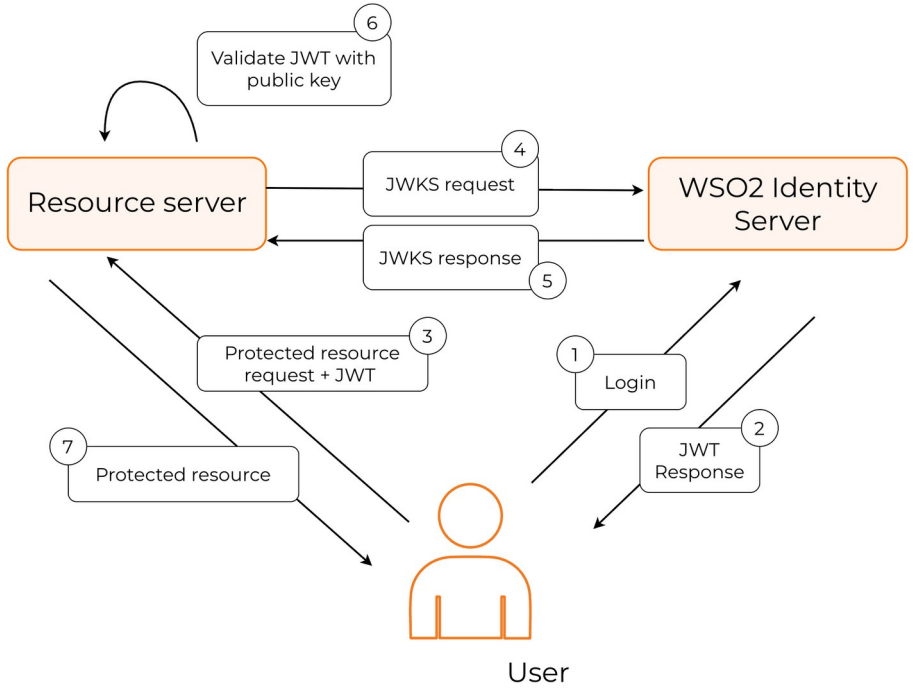
Cookies

- Cookie attributes
  - HttpOnly
    - JavaScript cannot read the cookie
    - Mitigates XSS
  - Secure
    - Cokie only sent over HTTPS
    - Required for production.
  - SameSite
    - Cookie behavior on cross-site requests.
    - Strict is most secure but can break legitimate flows
    - Lax is the balanced default
    - None requires Secure and is needed for cross-origin authenticated requests.
  - Domain and Path
    - Scope the cookie to the right URLs.



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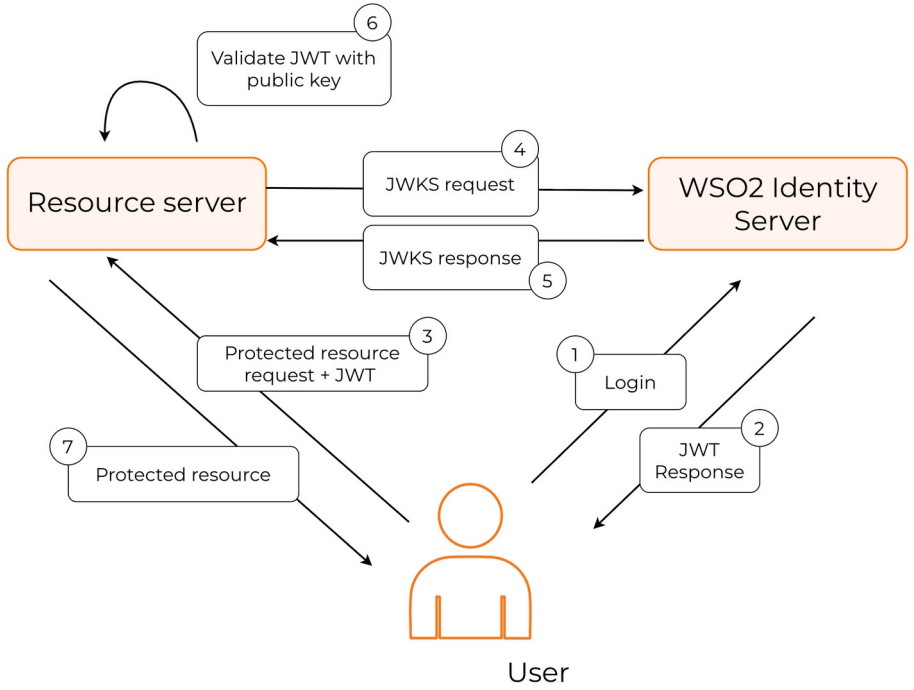
CSRF Revisited

- CookieCsrfTokenRepository
  - Stores CSRF token in a cookie readable by JavaScript
  - Sufficient when
    - SPA + BFF + same-origin
    - Trust the JavaScript on the page
  - Not sufficient when
    - XSS bypasses the same-origin assumption
    - User has a malicious browser extension
    - CSRF token is somehow accessible to the attacker



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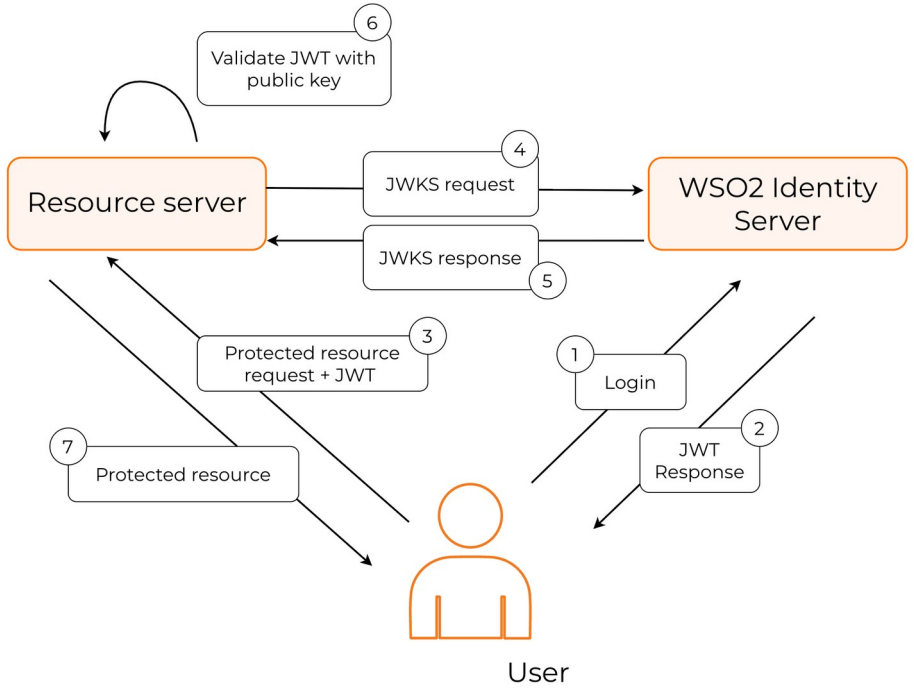
CSRF Revisited

- CookieCsrfTokenRepository
  - Compensating controls
    - Content Security Policy
    - Subresource integrity
    - Browser extension policies
- CSRF is one layer of defense, not the only layer



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Questions?